Measurement Error in Subjective Beliefs

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Measurement error in beliefs

Subjective expectations data is notoriously noisy:

- Blass, Lach, and Manski 2010; Juodis and Kucinskas 2023
- Question interpretation, rounding error, difficulty with probabilities, etc.

Importantly, error may be non-classical:

- Partisanship, reporting bias
- Survey fatigue, introspection

Error will affect precision and bias estimates...

But many papers in the expectations literature still do not address this (Delavande and Zafar 2019, Coibion et al. 2023, Cong et al. 2024)

This paper

Use an *information experiment* to estimate (and control for) measurement error:

1. Control group provides sequential measurements of the same belief (test-retest)
   $\rightarrow$ classical ME
2. Similar "types" of beliefs share parameters of measurement error process
   $\rightarrow$ relax classical ME

Can be useful in multiple ways:

- Correcting for bias in treatment effects
- Under assumption of *stability*, lets us study learning and other out-of-sample effects

Information experiments

Information experiments increasingly common (Fuster and Zafar 2023), allow researchers to:

1. Improve outcomes through by decreasing biases and information frictions
2. Introduces exogeneity and helps identify models (even when #1 fails)
3. **Now**: flexible model of measurement error (even when #1 and #2 fail)

Information experiments

Information experiments increasingly common (Fuster and Zafar 2023), allow researchers to:

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3. **Now**: flexible model of measurement error (even when #1 and #2 fail)

Specific type of experiment: same set of questions repeated before and after a treatment within the same survey session ("pseudo-panel data")

- **Pre- and post**: rules out Aloud et al. 2026
- **Same questions**: rules out Coibion et al. 2023
- **Same session**: rules out Jarke-Neuert, Perino, and Schwickert 2023

Costly + time-intensive, but useful.

Prior approaches

- Sensitivity tests: Wiswall and Zafar 2015, Delavande, Koşar, and Zafar 2025
- Classical ME (test-retest): Diaz-Serrano and Nilsson 2022
- IV: Armona, Fuster, and Zafar 2019
- SIMEX: Armona, Fuster, and Zafar 2019
- Probabilities, rounding and imprecision: Giustinelli, Manski, and Molinari 2022

Empirical application

Will illustrate using data from Armona, Fuster, and Zafar 2019, ReStud (AFZ):

- Information experiment on housing price growth (module in the NY Fed's SCE)
- Key beliefs: expected ZIP code housing price growth at $+1$ year and $+5$ years
- Treatment: historical information (control, past year, past 5 years)
- Beliefs elicited pre-and-post in the survey and then followed up after two months

Will use the data to:

1. Establish the need to address ME
2. Estimate ME parameters
3. Examine sensitivity of a few downstream parameter estimates

TABLE 1
Sample characteristics

	Full sample	Control	T1	T5	*P*-value[a]	Follow-up[b]
	(1)	(2)	(3)	(4)	(5)	(6)
Observations	1020	336	345	339		729
Age (in years)	50.4	51.1	50.7	49.4	0.301	51.5
	(15.3)	(15.7)	(15.1)	(15.1)		(15.2)
Male	54.3%	57.4%	58.0%	47.5%	0.008	53.9%
White and Non-hispanic	78.8%	78.6%	79.7%	78.2%	0.887	80.1%
Married	67.8%	65.8%	69.9%	67.8%	0.523	66.3%
Homeowner	74.3%	75.0%	73.6%	74.3%	0.919	75.4%
Tenure in town/city (in yrs)	19.1	19.3	19.6	18.3	0.548	19.5
	(16.5)	(16.4)	(17.1)	(16.0)		(16.8)
Bachelor's degree or more	55.3%	54.5%	54.2%	57.2%	0.680	55.0%
HH Income < $60,000	52.9%	51.5%	53.0%	54.3%	0.768	51.9%
HH Income < $30,000	20.0%	19.9%	20.3%	19.8%	0.985	20.2%
High numeracy[c]	73.5%	75.9%	73.3%	71.4%	0.413	74.2%
Employed	66.5%	66.1%	66.4%	67.0%	0.970	65.6%
Unemployed	3.6%	4.8%	3.5%	2.7%	0.338	3.4%
Not in the labour force	28.9%	28.3%	30.1%	28.3%	0.828	30.2%
Census region location:						
Northeast	15.8%	14.0%	16.5%	16.8%	0.542	16.5%
Midwest	21.4%	22.3%	20.0%	21.8%	0.738	21.3%
South	38.4%	37.2%	40.3%	37.8%	0.677	37.4%
West	24.4%	26.5%	23.2%	23.6%	0.553	24.8%

Table of Contents

1 Introduction

2 Basic model

3 Categories

4 Application

5 Conclusion

Two sequential measurements

Two sequential measurements of latent belief Y_i^*, $Y_{i,j}$, $j \in \{0,1\}$ pre- and post-treatment

- Use **control group** (pure control arm + treatments which won't affect Y_i^*):

$$Y_{i,0} = Y_i^* + \epsilon_{i,0}$$
$$Y_{i,1} = \alpha_1 + \beta_1 Y_i^* + \epsilon_{i,1}, \quad \mathbb{E}[\epsilon_{i,j}] = 0, \ \forall j$$

- Implicit normalization/anchoring: $\alpha_0 = 0$, $\beta_0 = 1$
- Also assume $\text{cov}(Y^*, \epsilon_j) = 0$

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Generates mean, variance, and covariance moments:

$$\mathbb{E}[Y_{i,j}] = \alpha_j + \beta_j \mu_{Y^*} \quad \forall j \quad \text{(2 conditions)} \quad \text{(2a)}$$
$$\text{var}(Y_{i,j}) = \beta_j^2 \sigma_{Y^*}^2 + \sigma_{\epsilon_j}^2 \quad \forall j \quad \text{(2 conditions)} \quad \text{(2b)}$$
$$\text{cov}(Y_{i,j}, Y_{i,j'}) = \beta_j \beta_{j'} \sigma_{Y^*}^2 + \text{cov}(\epsilon_{i,j}, \epsilon_{i,j'}) \forall j \neq j' \quad \text{(1 condition)} \quad \text{(2c)}$$

Two sequential measurements

Make additional assumptions for classical measurement error:

1. Unbiased measurements: $\alpha_j = 0$, $\beta_j = 1$, $\forall j$
2. Uncorrelated errors: $\text{cov}\left(\epsilon_{i,j}, \epsilon_{i,j'}\right) = 0$

Then:

1. Eqs. (2a) $\rightarrow$ latent mean μ_{Y^*}
2. Eq. (2c) $\rightarrow$ latent variance $\sigma^2_{Y^*}$
3. Eqs. (2b) $\rightarrow$ error variances $\sigma^2_{\epsilon_j}$

Should we worry?

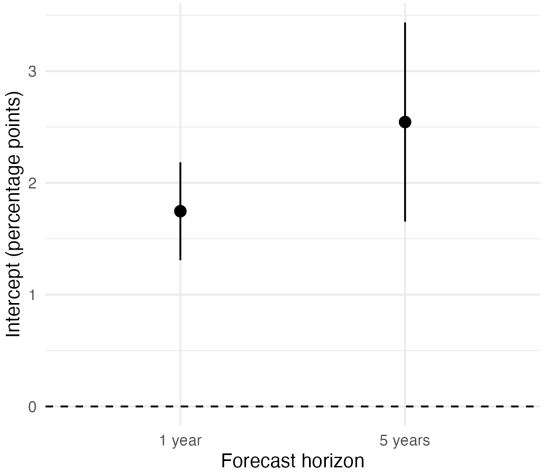
Can regress posterior on the prior:

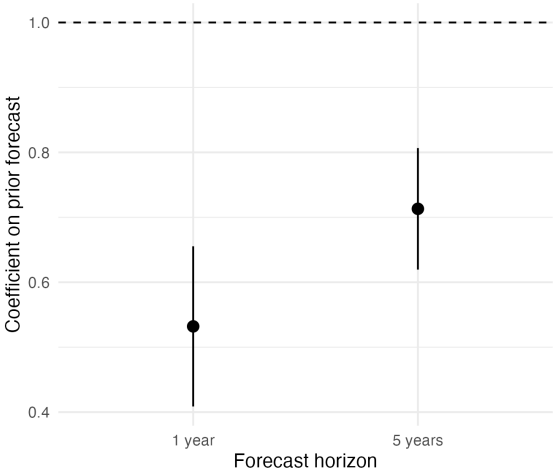
$$Y_{i,1} = \eta_0 + \eta_1 Y_{i,0} + u_i$$

If $\text{var}(\epsilon_{i,0}) = 0$, $\alpha_1 = 0$, $\beta_1 = 1$ then $\eta_0 = 0$, $\eta_1 = 1$

Turn to the AFZ data: expected growth in home prices at $+1$ and $+5$ years

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Turn to the AFZ data: expected growth in home prices at $+1$ and $+5$ years

- We see $\hat{\eta}_0 > 0$ and $\hat{\eta}_1 < 1$
- Compatible with classical measurement error or systematic pre-to-post deviations
- But which is going on matters!

Treatment effects

Simple constant treatment effect:

$$Y_i^*(D_i) = Y_i^* + \delta D_i$$

Treatment effect is $E[Y_{i,1} \mid D_i = 1] - E[Y_{i,1} \mid D_i = 0] = \beta_1 \delta$; biased if $\beta_1 \neq 1$

Undesirable in our context:

- **Survey fatigue**: $\mathbb{E}[|Y^* - Y_{i,1}|] > \mathbb{E}[|Y^* - Y_{i,0}|]$
- **Introspection**: $\mathbb{E}[|Y^* - Y_{i,1}|] < \mathbb{E}[|Y^* - Y_{i,0}|]$

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We will now try to relax the restrictions.

Table of Contents

1 Introduction

2 Basic model

3 Categories

4 Application

5 Conclusion

With two variables

Two separate latent beliefs indexed by $k \in \{1, 5\}$:

- Current application: expected housing price growth at $+1$ and $+5$ years
- Alternatively: expected tuition costs for oneself and for the "typical" person

Assume that $Y^*_{i,k}$ and $Y^*_{i,k'}$ share a *measurement system*:

$$Y_{i,k,j} = \alpha_{k,j} + \beta_j Y^*_{i,k} + \epsilon_{i,k,j} \tag{3}$$

β_j is shared – and will make some additional assumptions

2-variable system

This introduces new covariance conditions:

$$\mathbb{E}[Y_{i,k,j}] = \alpha_j + \beta_j \mu_{Y_k^*} \quad \forall k, j \quad (4 \text{ cond.}) \quad (4a)$$

$$\text{var}(Y_{i,k,j}) = \beta_j^2 \sigma^2_{Y_k^*} + \sigma^2_{\epsilon_{k,j}} \quad \forall k, j \quad (4 \text{ cond.}) \quad (4b)$$

$$\text{cov}(Y_{i,k,j}, Y_{i,k,j'}) = \beta_j \beta_{j'} \sigma^2_{Y_k^*} + \text{cov}(\epsilon_{i,k,j}, \epsilon_{i,k,j'}) \quad \forall j \neq j' \quad (2 \text{ cond.}) \quad (4c)$$

$$\text{cov}(Y_{i,k,j}, Y_{i,k',j}) = \beta_j^2 \text{cov}(Y_k^*, Y_{k'}^*) + \text{cov}(\epsilon_{i,k,j}, \epsilon_{i,k',j}) \quad \forall k \neq k' \quad (2 \text{ cond.}) \quad (4d)$$

$$\text{cov}(Y_{i,k,j}, Y_{i,k',j'}) = \beta_j \beta_{j'} \text{cov}(Y_k^*, Y_{k'}^*) + \text{cov}(\epsilon_{i,k,j}, \epsilon_{i,k',j'}) \quad \forall (k,j) \neq (k',j') \quad (2 \text{ cond.}) \quad (4e)$$

From **5** moment conditions to **14**:

- Also introduces new covariance parameters: 18 total
- So, still need assumptions – and attention to mapping from conditions to parameters

Identification

Assumptions on $\epsilon_{i,j,k}$:

1. ✗ cross-round, within-variable correlation: $cov(\epsilon_{i,j,k}, \epsilon_{i,j',k}) = 0, \forall k$
2. ✗ cross-round, cross-variable correlation: $cov(\epsilon_{i,j,k}, \epsilon_{i,j',k'}) = 0$
3. Shared within-round, cross-variable corr: $\text{corr}(\epsilon_{i,j,k}, \epsilon_{i,j,k'}) = \text{corr}(\epsilon_{i,j',k}, \epsilon_{i,j',k'}) \equiv \rho_{\epsilon,1,5}$

Leaves 13 parameters: $\theta = \left[\alpha_{1,k}, \beta_1, \mu_{Y_k^*}, \sigma^2_{Y_k^*}, \sigma_{Y_k^*, Y_{k'}^*}, \sigma^2_{\epsilon_{j,k}}, \rho_{\epsilon,1,5}\right]$, $\forall j, k$ are identified by:

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Leaves 13 parameters: $\theta = \left[\alpha_{1,k}, \beta_1, \mu_{Y^*_k}, \sigma^2_{Y^*_k}, \sigma_{Y^*_k, Y^*_{k'}}, \sigma^2_{\epsilon_{j,k}}, \rho_{\epsilon,1,5}\right]$, $\forall j, k$ are identified by:

- 2 covariance conditions from (4d) identify $\sigma_{Y^*_k, Y^*_{k'}}$ and β_1
- 4 mean conditions in (4a) pin down $\mu_{Y^*_k}$, $\alpha_{1,k}$
- 2 auto-covariance conditions in (4c) identify $\sigma^2_{Y^*_k}$ and $\sigma^2_{Y^*_{k'}}$
- 4 variance conditions in (4b) pin down $\sigma^2_{\epsilon_{k,j}}$
- 2 covariance conditions from (4e) serve as over-identifying restrictions on $\sigma_{Y^*_k, Y^*_{k'}}$ and β_j

Table of Contents

1 Introduction

2 Basic model

3 Categories

4 Application

5 Conclusion

Baseline results

Parameter	Estimate	Bootstrap SE
μ_1	3.353	(0.197)
μ_5	9.997	(0.502)
$\sigma^2_{H,1}$	9.360	(1.987)
$\sigma^2_{H,5}$	73.463	(13.758)
$\rho_{H,1,5}$	0.705	(0.058)
$\sigma^2_{\epsilon,1,0}$	3.697	(1.365)
$\sigma^2_{\epsilon,5,0}$	7.490	(9.199)
$\sigma^2_{\epsilon,1,1}$	6.377	(1.306)
$\sigma^2_{\epsilon,5,1}$	26.791	(7.620)
$\rho^{\text{pre}}_{\epsilon,1,5} = \rho^{\text{post}}_{\epsilon,1,5}$	0.410	(0.065)
$\alpha_{1,1}$	0.973	(0.475)
$\alpha_{1,5}$	2.021	(1.436)
β_1	0.783	(0.137)

Treatment effect

Now also index latent beliefs pre-and-post to reflect a treatment:

$$Y^*_{i,1,k} = Y^*_{i,0,k} + \sum_{s\in\{1,5\}} \delta_{s,k} D_{i,s} \left(Y^*_{i,0,s} - \tilde{Y}_{i,s} \right)$$

With $\tilde{Y}_{i,k}$ a signal about the true value of $Y^{oracle}_{i,k}$

$\rightarrow$ Housing price growth over past year/5 years gives information about the future

Naive approach: plug in reports and estimate:

$$\underbrace{Y_{i,1,k} - Y_{i,0,k}}_{\equiv \Delta_{i,k}} = \sum_{s\in\{1,5\}} \delta_{s,k} D_{i,s} \underbrace{\left(Y_{i,0,s} - \tilde{Y}_{i,s} \right)}_{\equiv G_{i,s}}$$

Bias

Consider three cases:

1. No measurement error: $\hat{\hat{\delta}}_{k,k} \to \delta_{k,k}$
2. Classical measurement error ($\alpha_{1,k} = 0$, $\beta_1 = 1$): direction of bias depends on $-(1 + \delta_{k,k}) \operatorname{var}(\epsilon_{i,0,k})$
   - If $\delta_{k,k} < -1$, upwards bias; understate effect
   - If $\delta_{k,k} = -1$, no bias
   - If $-1 < \delta_{k,k} < 0$, downwards bias; overstate effects
   - If $\delta_{k,k} = 0$, downwards bias; spurious negative effect
   - If $0 < \delta_{k,k}$, downwards bias; understate effects
3. Measurement error with $\alpha_{1,k} \neq 0$, $\beta_1 \neq 1$
   - Bias also depends on β_1 and $G_{i,s}$
   - Even if we instrument for $G_{i,s}$, $\hat{\hat{\delta}}_{k,k} \to \beta_1 \delta_{k,k}$

Estimating the treatment effect

Jointly estimate four new $\delta_{s,k}$, $s, k \in \{1, 5\}$ with the the measurement system

Additional mean, variance and covariance moments based on $Y_{i,j,k}$ and gaps $G_{i,s}$ for the treated groups

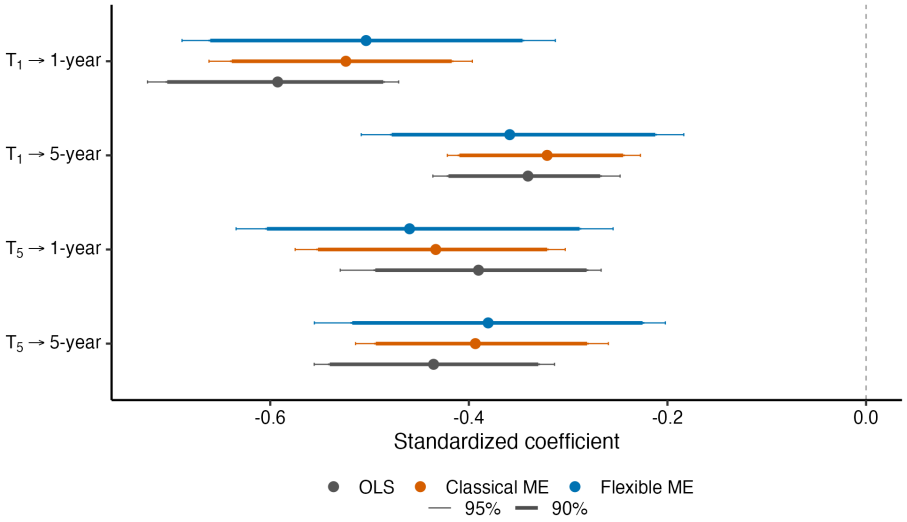
Matching 8 new moments; within treatment arm s:

$$\mathbb{E}[Y_{i,1,k}|D_{i,s} = 1] = \alpha_{1,k} + \beta_1 \left(\mathbb{E}_s[Y_{i,0,k}] + \delta_{s,k}\mathbb{E}_s[G_{i,s}]\right) \tag{5a}$$

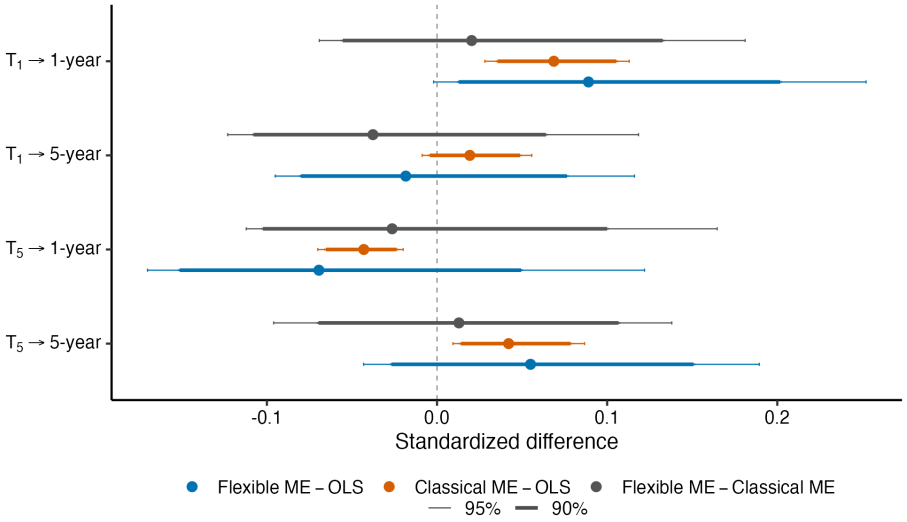
$$\begin{aligned}\text{cov}_s(G_{i,s}, Y_{i,1,k}) = \beta_1 \Big[&\text{cov}_s(G_{i,s}, Y_{i,0,k}) - \text{cov}(\epsilon_{i,0,s}, \epsilon_{i,0,k}) \\ &+ \delta_{s,k} \left(\text{var}_s(G_{i,s}) - \sigma_{\epsilon_{0,s}}\right) \Big]\end{aligned} \tag{5b}$$

Correction term

Treatment effect



Treatment effect differences



Are treatment effects persistent?

AFZ collect the same expectations two months later, $Y_{i,2,k}$

Assumption: $Y^*_{i,2,k}$ follows same measurement system as $Y^*_{i,k}$, e.g.

$$Y_{i,2,k} = Y^*_{i,2,k} + \epsilon_{i,2,k}$$

Where $\sigma^2_{\epsilon_{2,k}} = \sigma^2_{\epsilon_{0,k}}$, $\text{corr}(\epsilon_{i,2,s}, \epsilon_{i,2,k}) = \text{corr}(\epsilon_{i,0,s}, \epsilon_{i,0,k})$, and there is no correlation in errors between survey waves.

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Allow for a trend and mean-0 idiosyncratic changes over the two months in control group:

$$Y^*_{i,2,k} = Y^*_{i,0,k} + \eta_k + v_{i,k}$$

For the treated:

$$Y^*_{i,2,k} = Y^*_{i,0,k} + \eta_k + v_{i,k} + \sum_{s \in \{1,5\}} \delta'_{s,k} D_{i,s} G^*_{i,s}$$

Are treatment effects persistent?

Take the ME system as estimated and find $\hat{\eta}_k$ and $\hat{\delta}'_{s,k}$:

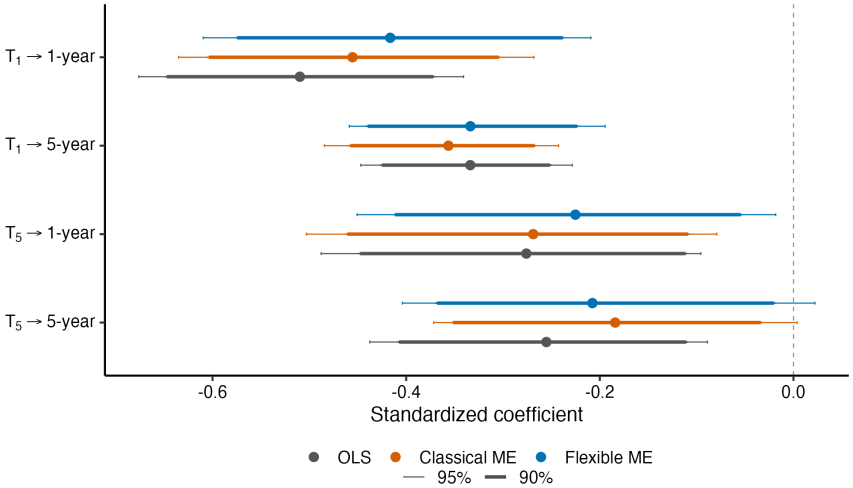
$$\underbrace{Y^*_{i,2,k} - Y^*_{i,0,k}}_{\equiv \Delta'_{i,k}} = \eta_k + \sum_{s \in \{1,5\}} \delta'_{s,k} D_{i,s} G^*_{i,s} + v_{i,k}$$

Idea:

1. Control group gives common baseline updating and variance of idiosyncratic changes (separate from variance of ME)
2. Treated groups give additional change based on original gap

Moments

Treatment effect



Treatment effect differences

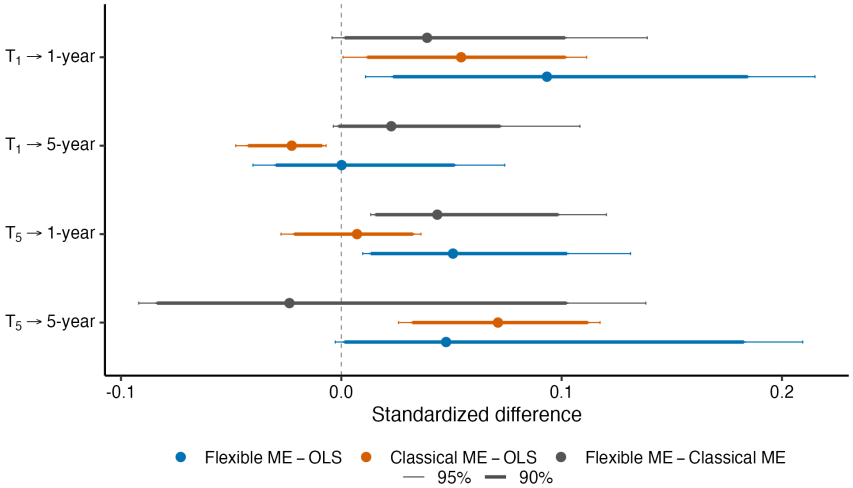


Table of Contents

1 Introduction

2 Basic model

3 Categories

4 Application

5 Conclusion

Conclusion

- Information treatment allows us to model measurement error in expectations data
- Critically: if multiple "types" of beliefs follow characteristics of the measurement system, can relax the classical ME assumptions
- Potentially important, since this may bias parameters of interest (treatment effects)

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Next steps:

1. Effect on estimates on downstream variables: the impact of expected housing price growth on investment decisions
2. And/or: alternate context with more bite
3. Test alternative assumptions and restrictions
4. Heterogeneity in observables or effort
5. Extend to elicited probabilities

Thank you

Correction term

We have the latent relationship:

$$\Delta Y^*_{i,k} = \delta_{s,k} G^*_{i,s},$$

so can express treatment effect as:

$$\delta_{s,k} = \frac{\text{cov}_s(G^*_{i,s}, \Delta Y^*_{i,k})}{\text{var}_s(G^*_{i,s})}.$$

Using our observed moments:

$$\delta_{s,k} = \frac{\text{cov}_s(G_{i,s}, Y_{i,1,k})/\beta_1 - \text{cov}_s(G_{i,s}, Y_{i,0,k}) + \text{cov}(\epsilon_{i,0,s}, \epsilon_{i,0,k})}{\text{var}_s(G_{i,s}) - \sigma^2_{\epsilon_{0,s}}}$$

Return

Persistence moments

From control group:

1. $\eta_k = \mathbb{E}\left[\Delta'_{i,k}\right]$
2. $\text{cov}\left(\Delta'_{i,k}, \Delta'_{i,s}\right) = \text{cov}\left(v_{i,k}, v_{i,s}\right) + 2\text{cov}\left(\epsilon_{i,0,k}, \epsilon_{i,0,s}\right)$

From treated group:

1. $\mathbb{E}[\Delta Y_{i,k} | D_{i,s} = 1] = \eta_k + \delta'_{s,k}\mathbb{E}[G_{i,s} | D_{i,s} = 1]$
2. $\text{cov}_s(G_{i,s}, \Delta Y_{i,k}) = \delta'_{s,k}\left[\text{var}_s(G_{i,s}) - \sigma^2_{\epsilon_{0,s}}\right] - \text{cov}\left(\epsilon_{i,0,k}, \epsilon_{i,0,s}\right)$

Return

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